

SAT Math Mastery · Week TP1J

SAT Trick Pack 1 — Part J

Time 120 minutes: 90 min solving + 30 min self-check and error log

Topics polygon angle sums · tangency · symmetry of a quadratic · unit chains · radical equations with a parameter · percent systems · fixed-plus-rate models · log vs line · proportional equations · difference of squares · grouping · exponent rewriting · conjugate factors · asymptotes

Name / Date _____

1 · SKILLS & IDEAS

- S1.** Equate a sum of angles to $180(n-2)$, solve for n , and reject roots that break convexity.
- S2.** Turn tangency into a zero discriminant; find every parameter value and the contact points.
- S3.** Use equal outputs of a quadratic to place the axis at a midpoint, then match it to the zeros.
- S4.** Chain unit conversions in the right direction; square (or cube) the factor for area (or volume).
- S5.** Handle rates that change between segments and convert once at the end.
- S6.** Substitute $t = \sqrt{x}$ and count valid roots with $t \geq 0$.
- S7.** Write each category as a fraction of the original total and eliminate variables.
- S8.** Isolate the per-unit part of a fixed-plus-rate model before applying a percent change.
- S9.** Compare a line with a logarithmic curve by splitting the domain at the intersections.
- S10.** Make one linear equation a multiple of another and read off the scale factor.
- S11.** Spot $A^2 - B^2$ and common binomial factors before expanding.
- S12.** Rewrite radicals as powers of one base; read bounds and attainment of exponential decay.

2 · REQUIRED FACTS

- The interior angles of a convex n -gon sum to $180(n-2)$ and each is less than 180° .
- A line is tangent to a parabola when the combined quadratic has discriminant 0; the contact has $x = -\frac{b}{2a}$.
- $f(p) = f(q)$ for a quadratic puts the axis at $x = \frac{p+q}{2}$, which is also the midpoint of the zeros.
- To convert squared units, square the linear factor ($1 \text{ yd} = 3 \text{ ft} \Rightarrow 1 \text{ yd}^2 = 9 \text{ ft}^2$).
- $\sqrt{x} = t$ forces $t \geq 0$; a quadratic in t with negative product of roots has exactly one root $t > 0$.
- Two equations $ax+by=c$ and $a'x+b'y=c'$ have infinitely many solutions when (a', b', c') is a constant multiple of (a, b, c) .
- $A^2 - B^2 = (A-B)(A+B)$; $a^m \cdot a^n = a^{m+n}$; $\sqrt[n]{a^m} = a^{m/n}$.
- A function $c + dr^x$ with $0 < r < 1$ and $d > 0$ decreases toward c but never reaches it.

3 · REQUIRED FORMULAS

$$\sum \theta = 180(n-2) = \frac{n}{2}(2a + (n-1)d), \quad \Delta = b^2 - 4ac = 0, \quad x_{\text{axis}} = \frac{p+q}{2}$$

$$A^2 - B^2 = (A-B)(A+B), \quad (u+v)(u-v) = u^2 - v^2, \quad \text{fare} = \text{fee} + \text{rate} \cdot d$$

4 · MINI THEORY SHEET

Parameter conditions. Translate the wording into one equation: sum of angles = $180(n-2)$, discriminant = 0, axis = midpoint, scale factor equal. Solve it completely, then test every root against the stated restrictions (convexity, $t \geq 0$, domain).

Units. Write the conversion as a fraction equal to 1 and make the unwanted unit cancel. For area use the squared factor. Convert at the end, not midway.

Expressions. Before expanding, look for a difference of squares or a shared binomial; substituting the given relation into the factored form is shorter and safer.

Functions. For a curve against a line, find the intersections, then test one point in every interval. For decay terms ask three separate questions: where does it head, is it ever reached, which way does it move?

Speed shortcuts.

Situation	Shortcut
angles in an arithmetic sequence	series sum = $180(n - 2)$; reject a root giving an angle $\geq 180^\circ$
line tangent to a parabola	discriminant = 0; contact at $x = -\frac{b}{2a}$; count both parameter values
equal outputs, zeros given	axis = midpoint of the inputs = midpoint of the zeros
mixed units in a trip	total each unit separately, convert once
map scale and area	convert each side, then square the factor for the area unit
$x - c\sqrt{x} = k$	$t = \sqrt{x}$; one valid root if the constant term is negative or the root is double
percent changes by category	fractions of the original total; subtract the total equation
fee plus rate, then rate changes	subtract the two data points for the rate; change only the rate
line against log curve	intersections first, then test a point in each region
infinitely many solutions	divide matching coefficients to get the multiple; apply it to the constants
$(3a + b)^2 - (a + 3b)^2$	factor as $(A - B)(A + B)$, then use the given relation
four-term expression and a given sum	group to expose the given binomial
roots and powers of one variable	everything to one base, then add exponents
$(Ax + B - \sqrt{c})(Ax + B + \sqrt{c})$	$= (Ax + B)^2 - c$; leading factor cancels in ratios
$c + dr^x, 0 < r < 1$	bound c is approached, never attained; function is monotone

5 · GUIDED EXAMPLES

Example 1 — Tangency: zero discriminant, both parameter values

The line $y = kx - 1$ is tangent to the parabola $y = x^2 + 2x + 3$. Find every possible k .

Set equal: $x^2 + (2 - k)x + 4 = 0$. Tangent means $\Delta = (2 - k)^2 - 16 = 0$, so $2 - k = \pm 4$ and $k = -2$ or $k = 6$. Both work.

Example 2 — Difference of squares before expanding

If $p^2 - q^2 = 15$, find $(2p + q)^2 - (p + 2q)^2$.

$$(2p + q - p - 2q)(2p + q + p + 2q) = (p - q) \cdot 3(p + q) = 3(p^2 - q^2) = 45.$$

Example 3 — Map scale, then square the factor

On a map, 1 cm represents 200 m. A lake is a 4 cm by 3 cm rectangle on the map. Find its real area in hectares (1 ha = 10,000 m²).

Real sides: 800 m and 600 m, area 480,000 m² = 48 ha. (Shortcut: $12 \text{ cm}^2 \times 200^2 = 480,000 \text{ m}^2$.)

6 · PRACTICE SET — 15 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. SAT CHALLENGE · NO CALCULATOR NEEDED

A convex polygon has n sides. Its interior angles, listed from smallest to largest, form an arithmetic sequence with smallest angle 120° and common difference 5° . How many sides does the polygon have?

- (A) 7 (B) 9
(C) 16 (D) 18

2. SAT CHALLENGE · NO CALCULATOR NEEDED

In the xy -plane, the line $y = kx + 4$, where k is a constant, is tangent to the parabola $y = x^2 + 4x + 13$ for exactly two values of k . For each of these values the line touches the parabola at one point. What is the greatest y -coordinate of these two points of tangency?

- (A) 10 (B) 13
(C) 34 (D) 3

3.

SAT CHALLENGE · NO CALCULATOR NEEDED

The function $g(x) = 2x^2 + bx + c$, where b and c are constants, has zeros $k + 1$ and $3k - 5$, where k is a constant. If $g(2) = g(10)$, what is the value of c ?

(A) 70

(B) 140

(C) 24

(D) 35

4.

SAT CHALLENGE · CALCULATOR

A car uses fuel at a rate of 6.0 liters per 100 kilometers on a highway and 10.0 liters per 100 kilometers in the city. On one trip the car travels 240 kilometers on the highway and 80 kilometers in the city. Using 1 mile = 1.6 kilometers and 1 gallon = 3.8 liters, which of the following is closest to the car's fuel efficiency for the whole trip, in miles per gallon?

(A) 87

(B) 6

(C) 30

(D) 34

5.

SAT CHALLENGE · NO CALCULATOR NEEDED

On a map, 1 inch represents 1,500 feet. A rectangular parking lot appears on the map as 2.4 inches by 1.6 inches. Which of the following is the actual area of the lot, in square yards? (1 yard = 3 feet)

(A) 8,640,000

(B) 320,000

(C) 960,000

(D) 2,880,000

6.

SAT CHALLENGE · NO CALCULATOR NEEDED

For a real number k , consider the equation

$$x - 4\sqrt{x} = k.$$

For how many integer values of k from -10 to 10 , inclusive, does the equation have exactly one real solution x ?

7.

SAT CHALLENGE · NO CALCULATOR NEEDED

A library's collection consists of fiction, nonfiction and reference books. Next year the number of fiction books will increase by 20%, the number of nonfiction books will decrease by 10%, and the number of reference books will not change. After these changes, the collection will contain 1% more books than it does now, and the number of fiction books will equal the number of nonfiction books. What percent of the current collection is reference books?

- (A) 35% (B) 65%
(C) 15% (D) 20%

8.

SAT CHALLENGE · CALCULATOR

A ride service charges a fixed pickup fee plus a constant rate for each mile traveled. A 5-mile ride costs \$14.50 and a 12-mile ride costs \$29.20. The service then raises its per-mile rate by 20% and leaves the pickup fee unchanged. Using 1 mile = 1.6 kilometers, what will a 20-kilometer ride cost after the change?

- (A) \$36.30 (B) \$54.40
(C) \$84.64 (D) \$35.50

9.

SAT CHALLENGE · DESMOS

Let $g(x) = \log_2 x$ for $x > 0$ and $h(x) = \frac{x+8}{6}$. For which values of x is $h(x) > g(x)$?

(A) $0 < x < 4$ or $x > 16$ (B) $4 < x < 16$ (C) $x > 16$ only(D) $x < 4$ or $x > 16$

10.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$3x + ky = 12$$

$$21x + 14y = 6m$$

In the system of equations above, k and m are constants. If the system has infinitely many solutions (x, y) , what is the value of km ?

11.

SAT CHALLENGE · NO CALCULATOR NEEDED

For real numbers a and b , $a^2 - b^2 = 36$. What is the value of $(3a + b)^2 - (a + 3b)^2$?

12.

SAT CHALLENGE · NO CALCULATOR NEEDED

If $2x + 3y = 8$, which of the following is equivalent to $2x^2 + 3xy - 4x - 6y$?

(A) $8x - 4y$ (B) $8x - 16$ (C) $8x - 8$ (D) $16 - 8x$

13.

SAT CHALLENGE · NO CALCULATOR NEEDED

Let t be a positive real number with $t \neq 1$. Define $m = \sqrt[3]{t^2}$ and $n = \sqrt{t}$. If

$$\frac{m^p \cdot n^3}{\sqrt[6]{t}} = t^5,$$

what is the value of p ?

14.

SAT CHALLENGE · NO CALCULATOR NEEDED

The function $f(x) = 2(3x + 1 - \sqrt{5})(3x + 1 + \sqrt{5})$ can be written as $f(x) = ax^2 + bx + c$, where a , b and c are constants. What is the value of $\frac{b+c}{a}$?

15.

SAT CHALLENGE · NO CALCULATOR NEEDED

Let $g(x) = 7 + 9\left(\frac{1}{4}\right)^x$ for all real numbers x . Consider the following statements.

I. $g(x) > 7$ for every real number x .

II. The minimum value of g is 7.

III. g is a decreasing function.

Which of the statements must be true?

(A) I and III only

(B) I and II only

(C) II and III only

(D) I, II, and III